

Yahui Li

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EDUCATION

Tsinghua University, Beijing, China

Aug. 2023 – Jul. 2027

B.S. in Theoretical and Applied Mechanics

GPA: 3.9/4.0

Tsien Excellence in Engineering Program

Relevant Courses: Viscous Fluid Mechanics, Solid Mechanics, Finite Element Method, Physics-based Simulation, Convex Optimization

RESEARCH EXPERIENCE

Harvard University

Cambridge, MA, USA

Research Intern, Advisor: Prof. Katia Bertoldi

Jul. 2026 - Present

Programmable Functional Responses Enabled by Sequential Buckling

- **Competing Buckling and Bulging Instabilities in Inflated Rubber Tubes:** Developed theoretical and finite-element models to investigate the sequence of global buckling and localized bulging in pressurized rubber tubes, aiming to identify parameter regimes governing buckle-first versus bulge-first responses.
- **Snap-Through Dynamics and Acoustics:** Simulated pressure-driven snap-through of arched shells in Abaqus using modal, fluid-cavity, and coupled Eulerian–Lagrangian (CEL) analyses. Characterized critical snap-through loads, post-buckled resonance frequencies, and transient fluid–structure interactions, and used the simulations to guide experiment design and identify geometries for broadband acoustic response.
- **Programmable Locomotion via Sequential Buckling:** Programmed sequential buckling by tailoring the geometry and spatial arrangement of arched shells to create distinct critical loads within a shared pneumatic cavity; characterized the resulting instability sequences experimentally and numerically for potential soft-robot locomotion.

Tsinghua University

Beijing, China

Research Intern, Advisor: Prof. Weixi Huang

Sep. 2025 - Jul. 2026

Flapping-enhanced active olfactory sampling in mosquitoes through near-field odor transport

First-author manuscript under review at [Bioinspiration & Biomimetics](#).

- **Numerical Method:** Extended an in-house Fortran flow solver by implementing passive scalar transport and a scalar jump-condition correction scheme for complex zero-thickness immersed boundaries.
- **Physical Discovery:** Analyzed flapping-induced near-field transport using Eulerian vorticity fields, Lagrangian particle tracking, and Lagrangian coherent structures (LCS), revealing that wing-generated vortices actively entrain odor toward the antennae and enhance olfactory sampling.
- **Physics-Guided Inverse Modeling:** Formulated odor-source localization as an inverse sensing problem and developed a physics-guided transfer-learning neural network combining a Gaussian plume prior with high-fidelity CFD data, improving localization accuracy over Gaussian-model inference alone.

University of Notre Dame

Notre Dame, IN, USA

ISURE Program, Advisor: Prof. Jianxun Wang

Jul. 2025 - Dec. 2025

Scientific Machine Learning for Turbulence Modeling and PDE Control

- **Turbulence Dimensionality Reduction:** Investigated conditional neural field models with low-dimensional latent representations for spatial dimension reduction of turbulence data.
[Contributed to a manuscript accepted in *Physics of Fluids*.](#)
- **Reinforcement Learning for PDE Control:** Implemented differentiable RL frameworks in Python and JAX for controlling nonlinear PDE systems, including the Kolmogorov–Sivashinsky equation, with the objective of suppressing unstable spatiotemporal fluctuations.
- **PDE Closure Modeling:** Explored learning-based closure models to reconstruct fine-scale dynamics from coarse numerical representations of nonlinear PDE systems.

Tsinghua University

Research Intern, Advisor: Prof. Hong Chen

Beijing, China

Sep. 2024 - Sep. 2025

Real-Time Event-Based Daily Human Action Recognition with On-Chip Learning on ANP-Is

First-author demo paper accepted at IEEE ISCAS 2026; a live demonstration presented in Shanghai, May 2026.

- **FPGA Design:** Developed Verilog modules for sequential and combinational logic, including handshake-based communication protocols for real-time event processing.
- **Hardware-Software Integration:** Integrated the FPGA with neuromorphic sensors and embedded computing platforms to build an end-to-end real-time action recognition pipeline.

PUBLICATIONS

- [1] **Y. Li**, J. Zhang, and W. Huang, “Flapping-Enhanced Active Olfactory Sampling in Mosquitoes through Near-Field Scalar Transport,” *Bioinspiration & Biomimetics*, under review.
- [2] **Y. Li**, J. Zhang, and H. Chen, “Live Demonstration: Real-Time Event-Based Daily Human Action Recognition with On-Chip Learning on ANP-I,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2026; accepted demo paper.
- [3] J. Guo, P. Du, X. Fan, **Y. Li**, and J. Wang, “Conditional Neural Field for Spatial Dimension Reduction of Turbulence Data: A Comparison Study,” *Physics of Fluids*, vol. 38, 2026.

SKILLS

Programming: Python, Fortran, JAX, C, MATLAB, Verilog

Simulation & Hardware: Abaqus, COMSOL, AutoCAD, Vivado

Experimental & Fabrication: PIV, Laser Cutting, 3D Printing

AWARDS

2023	Excellent Presentation Award , 13th Creative Competition & 2023 Tsinghua Student Creative Challenge	<i>Tsinghua University</i>
2024	Comprehensive Excellence Scholarship , Department-level Merit-based Scholarship	<i>Tsinghua University</i>
2024, 2025	Tsinghua Technology Taihu Scholarship , awarded for outstanding achievement in science and technology innovation	<i>Tsinghua University</i>